

Numerical modelling and design of edge-stiffened perforated cold-formed high-strength steel channel columns

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Keywords: High-Strength Steel, Finite element modelling, Edge-stiffened, Perforation.

Abstract. Cold-formed steel (CFS) channel sections are increasingly utilized as load-bearing components in modern building structures due to their structural efficiency and ease of fabrication. These sections often incorporate web perforations to facilitate the integration of services such as electrical and signal wiring, plumbing, air-conditioning systems, structural optimization, and maintenance access [1]. Conventionally, these perforations are created by punching and are left unstiffened, which limits their size and spacing due to the resulting reductions in structural performance [2]. Recently, researchers have addressed this issue by introducing edge-stiffening around the perforations, and interest in this approach is growing. Concurrently, significant advancements in steel manufacturing technologies have enabled the mass production of high-strength steel (HSS) with nominal yield strengths exceeding 460 MPa [3]. The superior mechanical properties of HSS, including increased yield strength and tensile capacity, make it highly suitable for long-span and high-rise structures, enabling reduced cross-sectional dimensions and improved connection performance. When edge-stiffening is combined with the use of HSS and optimized metal forming techniques, the structural efficiency of perforated CFS channel sections can be significantly enhanced. Despite these advantages, limited research has been conducted on the compressive behavior of edge-stiffened, perforated HSS channel sections. To address this gap, the present study undertakes a comprehensive investigation using validated numerical models to assess their axial load performance. The findings indicate that current international design standards do not fully capture the behavior of these advanced sections, highlighting the need for revised and improved design provisions.

References

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